

Inflation

Inflation

PYQs asked from this chapter:

Q1. The base year of All-India WPI has been revised from 2004-05 to: 2011-12

Q2. Stagflation means - High inflation and unemployment

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○ **Inflation:** The concept of Inflation refers to a sustained rise in the general level of prices of goods and services in an economy over a period of time. For a layman, inflation is just price rise.

Inflation leads to fall in purchasing power. When the price level rises, each unit of currency buys fewer goods and services; consequently, inflation is also erosion in the purchasing power of money — a loss of real value in the internal medium of exchange and unit of account in the economy.

A chief measure of price inflation is the inflation rate, i.e. the annualized percentage change in a general price index over time.

A rise in the general level of prices; a sustained rise in the general level of prices; persistent increases in the general level of prices; an increase in the general level of prices in an economy that is sustained over time; rising prices across the board—is inflation. These are some of the most common academic definitions of inflation. When the general level of prices is falling over a period of time this is deflation, the opposite situation of inflation. It is also known as disinflation. But in contemporary economics, deflation or disinflation not used to indicate fall in prices. Instead, a price rise is termed a 'rise in inflation' and a price fall is termed a 'fall in inflation'. As instruments of deflation, the policy includes fiscal measures (as for example, tax increase) or monetary measures (as for example, increase in interest rate).

➤ Effect of inflation:

Inflation has both positive and negative effects on an economy. Negative effects of inflation include loss in stability in the real value of money and other monetary items over time, uncertainty about future inflation may discourage investment and saving, and high inflation may lead to shortages of goods if consumers begin hoarding out of concern that prices will increase further in the future. Positive effects include a mitigation of economic recessions and debt relief by reducing the real level of debt.

❖ **CAUSES OF INFLATION:** There are many factors that can trigger inflationary pressure in an economy. The most important of these are:

◆ **Demand-Pull Inflation:** It is a rise in general prices caused by increasing aggregate demand for goods and services. Increasing quantity of money in the hands of the people increases the aggregate demand for goods and services, and if aggregate supply does not follow the suit, prices rise. (Demand exceeding supply leads to shortage and hence, an increase in price). This was a Keynesian idea.

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♦ **Cost-Push Inflation:** It is a type of inflation caused by substantial increases in the cost of **production of important goods or services** where no suitable alternative is available. For example, if **prices of some key inputs like oil rise**, producers will have to either adjust output supply or translate the higher costs into higher output prices. When output declines because of cost pressure on producers, there will be a shortage in output markets and as a result prices will rise.

❖ **Inflation Control measures:** From the above-given reasons for inflation and the measures to control it, In practice, governments around the world distance themselves from this debate and have been taking recourse to all possible options while controlling inflation. The governments resort to the following options to check rising inflation:

(i) As a **supply side** measure, the government may go for **import of goods** which are in short-supply—as a short-term measure (as happened in India in the case of ‘onion’ and meeting the buffer stock norm of wheat). As a long-term measure, governments go on to **increase the production** to matching the level of demand. **Storage, transportation, distribution, hoarding** are the other aspects of price management of this category.

(ii) As a **cost side measure**, governments may try to cool down the price by cutting down the production cost of goods showing price rise with the **help of tax breaks**—cuts in the excise and custom duties (as happened in June 2003 in India in the case of crude oil and steel). This helps as a short-term measure. In the long-term, **better production process, technological innovations** etc., are helpful. Increasing income of the people is the monetary measure to avoid the heat of such inflation.

(iii) The governments may take recourse to **tighter monetary policy** to cool down either the demand-pull or the cost-push inflations. This is basically intended to cut down the money supply in the economy by **siphoning out the extra money** (as RBI increases the Cash Reserve Ratio of banks in India) from the economy and by making money costlier (as RBI increases the Bank Rate or Repo Rate in India). This is a short-term measure. In the long-run, the best way is to increase production with the help of the best production practices.

➤ **TYPES OF INFLATION:** Depending upon the range of increase, and its severity, inflation may be classified into three broad categories:

1. **Low Inflation:** Such inflation is slow and on predictable lines, which might be called small or gradual. This is a comparative term which puts it opposite to the faster, bigger and unpredictable inflations. Low inflation takes place in a longer period and the range of increase is **usually in ‘single digit’**. Such inflation has also been called as **‘creeping inflation’**. We may take an example of the monthly inflation rate of a country for six months being 2.3 per cent, 2.5 per cent, 2.7 per cent, 2.9 per cent, 3.1 per cent and 3.3 per cent. Here the range of change is of 1.0 per cent and over a period of six months.

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2. Galloping Inflation: This is a 'very high inflation' running in the range of double-digit or triple digit (i.e., 20 per cent, 100 per cent or 200 per cent in a year). In the decades of 1970s and 1980s, many Latin American countries such as Argentina, Chile and Brazil had such rates of inflation—in the range of 50 to 700 per cent. The Russian economy did show such inflation after the disintegration of the ex-USSR in the late 1980s. Contemporary journalism has given some other names to this **inflation hopping, inflation, jumping, inflation and running or runaway inflation**.

3. Hyperinflation: This form of inflation is 'large and accelerating' which might have the annual rates in million or even trillion. In such inflation not only the range of increase is very large, but the increase takes place in a very short span of time, prices shoot up overnight. Some recent examples of hyperinflation had been the Bolivian inflation of mid-1985 (24,000 per cent per annum) and the Yugoslavian inflation of 1993 (20 per cent per day).

➤ **OTHER VARIANTS OF INFLATION:** Other than the three broad categories we analyzed above, some other variants of inflation are also considered by governments in their policymaking:

I. Bottleneck Inflation: This inflation takes place when the supply falls drastically and the demand remains at the same level. Such situations arise due to supply-side accidents, hazards or mismanagement which is also known as 'structural inflation'.

II. Core Inflation: This nomenclature is based on the inclusion or exclusion of the goods and services while calculating inflation. Popular in western economies, core inflation shows price rise in all goods and services excluding energy and food articles. In India, it was first time used in the financial year 2000–01 when the government expressed that it was under control—it means the prices of manufactured goods were under control.

This was criticized by experts on account of excluding food articles and energy out of the inflation and feeling satisfied on the inflation front. Basically, in the western economies, food and energy are not the problems for the masses, while in India these two segments are of most vital importance for the common people.

III. Reflation: Reflation is a situation often deliberately brought by the government to reduce unemployment and increase demand by going for higher levels of economic growth. Governments go for higher public expenditures, tax cuts, interest rate cuts, etc. Fiscal deficit rises, extra money is generally printed at higher level of growth, wages increase and there is almost no improvement in unemployment.

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Reflation can also be understood from a different angle—when the economy is crossing a cycle of recession (low inflation, high unemployment, low demand, etc.) and government takes some economic policy decisions to revive the economy from recession, Governments go for higher public expenditures, tax cuts, interest rate cuts, wage increase etc. then certain goods see sudden and temporary increase in their prices, such price rise is also known as reflation.

IV. Stagflation: Stagflation is a situation in an economy when inflation and unemployment both are at higher levels, contrary to conventional belief. Such a situation first arose in the 1970s in the US economy and in many Euro-American economies. When the economy is passing through the cycle of stagnation (i.e., long period of low aggregate demand in relation to its productive capacity) and the government shuffles with the economic policy, a sudden and temporary price rise is seen in some of the goods—such inflation is also known as stagflation. Stagflation is basically a combination of high inflation and low growth.

V. Deflation: When the general level of prices is falling over a period of time this is deflation, the opposite situation of inflation. Or when inflation falls below zero, the economy is said to be in a state of deflation.

VI. Disinflation: Disinflation is a situation when the rate of inflation tends to fall over time but remains positive. Put in the form of an example, deflation would be an inflation rate of -1%, while disinflation would be a change in the inflation rate from 3% one year to 2% in the next year.

VII. Creeping Inflation: When the rise in prices is very slow (less than 3% per annum) like that of a snail or creeper, it is called creeping inflation. Such an increase in prices is regarded safe and essential for economic growth.

VIII. Walking Inflation: When prices rise moderately and the annual inflation rate is a single digit (3% - 10%), it is called walking or trotting inflation. Inflation at this rate is a warning signal for the government to control it before it turns high.

❖ OTHER IMPORTANT TERMS:

Inflationary Gap: The excess of total government spending above the national income (i.e., fiscal deficit) is known as inflationary gap.

Deflationary Gap: The shortfall in total spending of the government (i.e., fiscal surplus) over the national income creates deflationary gaps in the economy. This is a situation of producing more than the demand and the economy usually heads for a general slowdown in the level of demand. This is also known as the output gap.

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Inflation Accounting: When a firm calculates its profits after adjusting the effects of current level of inflation, this process is known as inflation accounting.

Inflation Premium: An inflation premium is the part of prevailing interest rates that results from lenders compensating for expected inflation by pushing nominal interest rates to higher levels. (Difference between existing rate and interest rate after inflation).

Phillips Curve: It is a graphic curve which advocates a relationship between **inflation and unemployment in an economy**. As per the curve there is an inverse relationship between them. The curve suggests that lower the inflation, higher the unemployment and higher the inflation, lower the unemployment.

Inflation Targeting: The announcement of an official target range for inflation is known as inflation targeting. It is done by the Central Bank in an economy as a part of their monetary policy to realise the objective of a stable rate of inflation (the Government of India asked the RBI to perform this function in the early 1970s).

India commenced inflation targeting 'formally' in February 2015 when an agreement between the GoI and the RBI was signed related to it—the Agreement on Monetary Policy Framework. The agreement provides the aim of inflation targeting in this way—"it is essential to have a modern monetary framework to meet the challenge of an increasingly complex economy. Whereas the objective of monetary policy is to primarily maintain price stability, while keeping in mind the objective of growth.

The major highlights of the agreement is as given below:

- **RBI will aim to keep inflation level in the range of 2-6%.**
- **Every six months, the RBI to publish a document explaining:**
 - a. Source of inflation;
 - b. Forecasts of inflation for the period between six to eighteen months from the date of the publication of the document; and
- **If the RBI fails to meet the target it shall set out in a report to the GoI:**
 - a. the reasons for its failure to achieve the target under set in this agreement;
 - b. remedial actions proposed to be taken by the RBI; and
 - c. an estimate of the time-period within which the target would be achieved pursuant to timely implementation of proposed remedial actions.
- **Any dispute regarding the interpretation or implementation of the agreement to be resolved between the Governor, RBI and the GoI.**

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➡ **Skewflation:** When there is a price rise of one or a small group of commodities over a sustained period of time, without a traditional designation.

➡ **GDP Deflator:** This is the ratio of GDP at Current Prices to GDP at Constant Prices. If GDP at Current Prices is equal to the GDP at Constant Prices, GDP deflator will be 1, implying no change in price level.

To understand GDP Deflator, consider an example where India is producing 10kg of wheat at the market price of Rs. 10 in 2012 and 11kg of wheat at Rs. 10.5 in 2013.

	<u>2012</u>	<u>2013</u>
Wheat	10kg × Rs. 10	11kg × Rs. 10.5

If we want to know the inflation from 2012 to 2013 then we can see that it is 5% (Rs. 10 to Rs. 10.5). But inflation can be calculated in another way also by keeping the quantity constant in both the years .

In the above example, Nominal GDP of 2013 = 11kg × Rs. 10.5 = Rs 115.5
and, Real GDP of 2013 = 11kg × Rs. 10 = Rs 110

If we divide Nominal GDP of 2013 by Real GDP of 2013, what we get is the change in price because in Nominal GDP and Real GDP of 2013, the quantity is same (constant) i.e. 11kg.

$$\begin{aligned} \text{So, Change in price from 2012 to 2013} &= \frac{\text{Nominal GDP of 2013}}{\text{Real GDP of 2013}} \\ &= \frac{11\text{kg} \times \text{Rs. } 10.5}{11\text{kg} \times \text{Rs. } 10} = \frac{\text{Rs } 115.5}{\text{Rs. } 110} = \frac{1.05}{1} \end{aligned}$$

Which means the prices have increased by 1.05 times i.e. inflation is 5%. And this ratio of Nominal GDP/ Real GDP is called the GDP deflator;

$$\text{So, GDP Deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}}$$

GDP deflator is published by **National Statistics Office (NSO), Ministry of Statistics and Programme Implementation (MoSPI)**.

The following are some basic differences in CPI, WPI and GDP deflator:

- In wholesale market services are not traded, so WPI does not include the inflation in services, while CPI and GDP deflator capture inflation in services also.
- The goods purchased by consumers in the retail market do not represent all the goods produced in the country (capital goods are purchased by the companies), so CPI does not

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include such capital goods but GDP deflator takes into account all such goods and services produced in the country.

- CPI and WPI include prices of goods produced domestically and imported both but GDP deflator does not include prices of imported goods.
- The weights are constant (in the basket) in CPI and WPI, but they differ according to production level of each good and services in GDP deflator.

➡ **BASE EFFECT:** It refers to the impact of the rise in price level (i.e., last year's inflation) in the previous year over the corresponding rise in price levels in the current year (i.e., current inflation).

For example –

Case 1 – The price index of January 2016 (base year) is 110 and that of January 2017 is 120.

Now Inflation of Jan 2017 = $(120-110)/110 * 100$ which comes out to be 9.09%.

Case 2 – The price index for March 2017 (base year) is 180 and that of March 2018 is 190. Now Inflation of March 2018 = $(190-180)/180 * 100$ which comes out to be 5.55%.

Now we see in both the case the increase in price index is 10 but the rate of inflation is different. This is due to the base effect.

➡ **Base effect in terms of GDP:**

In 2019-20, India's GDP was at ₹145.7 lakh crore and ₹134.4 lakh crore in 2020-21 and in 2021-22 ₹149.2 lakh crore.

The 11 per cent growth in 2021-22 on the lower base of ₹134.4 lakh crore in 2020-21 have taken the GDP to ₹149.2 lakh crore.

This means the GDP is just 2.4% more than where it was in 2019-20 and the 11 per cent growth will be due to the abnormal contraction in 2020-21 owing to the coronavirus pandemic. That's how the base effect will play its part in the expected growth.

➤ **MEASURES OF INFLATION:** The rate of inflation is measured on the basis of price indices which are of two kinds—**Wholesale Price Index (WPI) and Consumer Price Index (CPI)**. A price index is a measure of the average level of prices, which means that it does not show the exact price rise or fall of a single good. The rate of inflation is the rate of change of general price level which is measured as follows:

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$$\text{Rate of inflation (year } x) = \frac{\text{Price level (year } x) - \text{Price level (year } x-1)}{\text{Price level (year } x-1)} \times 100$$

This rate shows up in percentage form (%), though inflation is also shown in numbers, i.e., digits. A **price index** is a weighted average of the prices of a number of goods and services. In the index the total weight is taken as 100 at a particular year of the past (the base year), this when compared to the current year shows a rise or fall in the prices of current year, there is a rise or fall in the '100' in comparison to the base year—and this inflation is measured in digits.

➡ **INFLATION IN INDIA:** Every economy calculates its inflation for efficient financial administration as the multi-dimensional effects of inflation make it necessary. India calculates its inflation on two price indices, i.e., the wholesale price index (WPI) and the consumer price index (CPI). While the WPI-inflation is used at the macro-level policymaking, the CPI-inflation is used for micro-level analyses. The inflation at the WPI is the inflation of the economy. Both the indices follow the 'point-to-point' method and may be shown in points (i.e., digit) as well as in percentage relative to a particular base year.

❖ **Wholesale Price Index:** The Office of the Economic Adviser in the Office of Economic Adviser, Department for Promotion of Industry and Internal Trade, Ministry of Commerce & Industry is responsible for compiling WPI and releasing it.

The first Economic Adviser to the Government of India in pre-Independence India, Sir Theodore E.G. Gregory (period 1937-1946) had started the 'Quick' series, using the week ended August 19, 1939 as base and computed the index from the week commencing January 10, 1942.

Six revisions have taken place introducing the new base years, viz., 1952-53, 1961-62, 1970-71, 1981-82, 1993-94 and 2004- 05.

The new series of WPI with base year of 2011-12 is the seventh revision of WPI.

WPI is used as a deflator of various nominal macroeconomic variables including Gross Domestic Product (GDP). The WPI based inflation estimates also serve as an important determinant, in formulation of trade, fiscal and other economic policies by the Government. WPI is also used for the purpose of escalation clauses in the supply of raw materials, machinery and construction work.

Wholesale Price Index (WPI) is measured by the Office of the Economic Adviser, Department for Promotion of Industry and Internal Trade is entrusted with the task of releasing this index.

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Wholesale Price Index (WPI) are released on the 14th of every month (or next working day) with a time lag of two weeks of the reference month.

➡ **New Series of WPI:** For the new series with base 2011-12=100, a Working Group was constituted on 19th March 2012 chaired by Late **Dr. Saumitra Chaudhuri, Member**, erstwhile Planning Commission and comprised most stakeholders. In light of the recommendations the government recently announced the New Series of Wholesale Price Index **with base 2011-12=100** and also added a new WPI food index to capture the rate of inflation in food items.

➡ **Features of the Revised Series of WPI:** In the updated WPI basket, the number of items has been increased to **697**.

The number of quotations are increased to **8331**, an increase by 2849 quotations (52%).

In the revised series, WPI will continue to constitute three major groups namely

- Primary article (117 items-22.62% weightage),
- Fuel & Power (16 items-13.15% weightage) and
- Manufacturing products (564 items-64.23% weightage).

In the new WPI series, elementary price index (i.e. at the item level) has been computed using the **geometric mean of the price relatives (Jevons' Index)** as opposed to the practice of taking arithmetic mean of price relatives (Carli Index) as was the case in the previous WPI series.

The index is compiled as per the methodology approved by the Technical Advisory Committee (TAC).

➡ **Technical Review Committee (TRC)** is chaired by Secretary, Department for Promotion of Industry and Internal Trade. This committee will meet at least once a year for inter-alia recommending methodology for improving the coverage and quality of WPI like identifying new items that need to be included in the item basket and removing those that have lost its relevance or are no longer being produced.

❖ **Consumer Price Index (CPIs):**

In India, segment specific Consumer Price Index (CPIs), namely CPI (IW), CPI (AL+RL), **CPI (Rural/Urban/Combined)** are being compiled regularly, catering to the need of specific population group. CPI (UNME) which has been discontinued w.e.f. December, 2010, was meant for urban non-manual employees (UNME).

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The following CPIs are being measured in India:

- CPI (IW)- Consumer Price Index for Industrial Workers
- ALL-INDIA CONSUMER PRICE INDEX NUMBERS FOR AGRICULTURAL AND RURAL LABOURERS
- CPI (Rural/Urban/Combined)

Of these, the **first two are compiled by the Labour Bureau in the Ministry of Labour and Employment**. Fourth is compiled by the National Statistical Office (NSO) in the Ministry of Statistics and Programme Implementation.

CPI-IW: (Revised base year: 2016=100): The Labour Bureau, an attached office of the M/o Labour & Employment, has been compiling Consumer Price Index for Industrial Workers every month on the basis of retail prices of 463 commodities collected from 317 markets spread over 88 industrially important centres in the country. The index is compiled for 88 centres and All-India and is released for every month on the last working day of succeeding month.

- The number of selected markets for collection of retail price data has also been increased to 317 markets
- The number of items directly retained in the index basket has increased to 463 items as against 392 items in the 2001 series.
- The number of States/UTs has increased to 28 under 2016 series as against 25 in the 2001 series.

In the new series, as per the direction of Technical Advisory Committee (TAC) on Statistics of Prices and Cost of Living (SPCL), the Geometric mean-based methodology (GM of Price Relatives) is used for compilation of indices as against Arithmetic mean used in 2001 series.

The index is compiled using weights derived from the Working-Class Family Income and Expenditure Survey.

The wages/salaries of the central government employees are revised on the basis of the changes occurring in this index, the dearness allowance (DA) is announced twice a year. When the Pay Commission recommends pay revisions, the base is the CPI (IW).

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CPI -IW group items:

Sr. No.	Groups
I	Food & Beverages (Highest weightage)
II	Pan, Supari, Tobacco & Intoxicants (least weightage)
III	Clothing & Footwear
IV	Housing (3rd highest weightage)
V	Fuel & Light
VI	Miscellaneous (2 nd highest weightage)

ALL-INDIA CONSUMER PRICE INDEX NUMBERS FOR AGRICULTURAL AND RURAL LABOURERS: (Base: 1986-87:100)

The Labour Bureau under MINISTRY OF LABOUR & EMPLOYMENT has been compiling CPI Numbers for Agricultural Labourers since September, 1964.

Consumer Price Index Numbers for Agricultural Labourers and Rural Labourers (CPI AL&RL) on base: 1986-87=100, which serve as a guiding factor for fixation and revision of minimum wages of labourers engaged in agricultural occupations under the Minimum Wages Act, 1948 are compiled, maintained and disseminated by the Labour Bureau.

These indices, besides being utilized for wage indexation are also utilized for revising cooking cost under Pradhan Mantri Poshan Shakti Nirman Scheme (PM POSHAN) , measuring the inflation and for research & policy making.

The Consumer Price Index Numbers for Agricultural Labourers (CPI-AL) and Rural Labourers (CPI-RL) on base 1986-87=100 are being compiled on monthly basis by utilizing the price data collected by the Field Operations Division of the National Statistical Office from **600 sample villages spread over 66 National Sample Survey (NSS) regions of 20 States and released on 20th of the succeeding month.**

The CPI-AL&RL are compiled as per the methodology approved by the Technical Advisory Committee on Statistics of Prices and Cost of Living (TAC on SPCL)

Rural Labour Households are those households whose income during the last 365 days from wage paid manual labour (agricultural and/or non-agricultural) was more than either from paid non-manual employment or from self- employment.

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Agricultural Labour household: The rural labour households, which derive 50 per cent or more of their total income from wage paid manual labour in agricultural activities, are treated as agricultural labour households.

➡ Components of CPI (AL&RL):

Group
General Index
Food
Pan, Supari, etc.
Fuel & Light
Clothing, Bedding & Footwear
Miscellaneous

➡ CPI (Rural/Urban/Combined):

CPI measures the average change in prices of fixed baskets of goods and services that households purchase for the purpose of consumption. It is also called *Retail Inflation*.

The National Statistical Commission (NSC), under **Dr. C. Rangarajan, in its Report (2001)** recommended the compilation of **CPI for rural and urban areas**.

It was in 2011 that the government announced a new Consumer Price Index (CPI) – CPI (Rural); CPI (Urban) and by combining them into a ‘national’ CPI-C (where ‘C’ stands for ‘Combined’).

The National Statistics Office (NSO), Ministry of Statistics and Programme Implementation has revised the Base Year of the Consumer Price Index (CPI) from 2010=100 to 2012=100.

The number of Groups, which was five in the old series, has now been increased to following six:

- Food and beverages (45.86% weightage combined of rural and urban CPI)
- Pan, tobacco and intoxicants (2.38% weightage combined of rural and urban CPI)
- Clothing and Footwear (6.53 % weightage combined of rural and urban CPI)
- Housing (10.07 % weightage combined of rural and urban CPI)
- Fuel and Light (6.84 % weightage combined of rural and urban CPI)
- Miscellaneous (28.32 % weightage combined of rural and urban CPI)

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'Pan, tobacco and intoxicants', which was a Sub-group under the group 'Food, beverages and tobacco', has now been made as a separate group. Accordingly, the group 'Food, beverages and tobacco' has been changed to 'Food and beverages'.

The elementary/item indices are now being computed using Geometric Mean (GM) of the Price Relatives of Current Prices with respect to Base Prices of different markets in consonance with the international practice.

The basket of items and their weighing diagrams have been prepared using the Modified Mixed Reference Period (MMRP) data of Consumer Expenditure Survey (CES), 2011-12, of the 68th Round of National Sample Survey (NSS).

The Reserve Bank of India (RBI) has started using CPI-combined as the sole inflation **measure for the purpose of monetary policy after the recommendation of Urjit Patel committee**. As per the agreement on Monetary Policy Framework between the Government and the RBI dated February 20, 2015 the sole objective of RBI is price stability and a target is set for inflation as measured by the Consumer Price Index-Combined.

WPI is released monthly and annually and also CPI is released monthly and annually.

CPI is being released with two weeks lag time

Core inflation is calculated by excluding the more volatile components like food and fuel from consumer price index (CPI)

Some official and nonofficial versions of the suitable range of inflation pointed out from time to time:

- (i) The **Chakravarty Committee** (1985) treated 4 per cent inflation as acceptable for the economy in its report on the monetary system. He also added that this level of price rise will facilitate the purpose of attracting investment for the desired level of growth.
- (ii) The Government of India accepted a range of 4 to 6 per cent inflation as acceptable for the economy citing the world average of 0 to 3 per cent at the time (1997-98).
- (iii) The RBI Governor C. Rangarajan advocated that inflation rate must come down initially to 6 to 7 per cent and eventually to 5 to 6 per cent on an average over the years.
- (iv) The **Tarapore Committee** on Capital Account Convertibility recommended an acceptable range of 3 to 5 per cent inflation for the three year period (1997-98 to 1999-2000).

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In February 2015, India formally commenced the process of 'inflation targeting'. Now, the new monthly CPI (C), is taken as the measure of **headline inflation** and is tracked by the RBI to anchor its monetary policy and the healthy annual range for it is between 2 to 6 per cent.

➡ **Headline inflation:** Headline inflation is a measure of the total inflation within an economy, including commodities such as food and energy prices.

INDEX OF INDUSTRIAL PRODUCTION: The all India index of Industrial Production (IIP) is a composite indicator that measures **the short-term changes in the volume of production** of a basket of industrial products during a given period with respect to that in a chosen base period.

It is compiled and published monthly by the **National Statistics Office (NSO)**, Ministry of Statistics and Programme Implementation.

The Quick Estimates of Index of Industrial Production (IIP) are released on 12th of every month (or previous working day if 12th is a holiday) **with a six weeks lag**.

The base year of the all-India Index of Industrial Production (IIP) has also been revised from 2004-05 to **2011-12** to not only reflect the changes in the industrial sector but to also align it with the base year of other macroeconomic indicators like the Gross Domestic Product (GDP), Wholesale Price Index (WPI).

Technical Review Committee, chaired by Secretary, Ministry of Statistics & PI. This Committee will meet at least once a year for identifying new items that need to be included in the item basket and removing those that have lost its relevance in the industrial sector or are no longer being produced.

IIP in the revised series will continue to represent the Mining, Manufacturing and Electricity sectors. The revised series uses the National Industrial Classification (NIC) 2008 for the purpose of classification of industrial production. The unit coverage of IIP will, as before, cover entities in the organized sector units registered under the Factories Act, 1948.

The items selected for new series of IIP with base 2011-12 comprised of **839 items with the highest share of Manufacturing items i.e 809**. Mining and Electricity sectors will be represented by a single item index.

The sectoral composition of the IIP is as follows:

- **Sector(weights%)**
- Mining (14.373%)
- Manufacturing (77.633%)
- Electricity (7.994%)

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The electricity sector now includes data from renewable energy sources.

✓ **Note: The Index of Eight Core Industries (BASE: 2011-12=100)** comprises 40.27 per cent of the weight of items included in the Index of Industrial Production (IIP). Coal, crude oil, natural gas, refinery products, Fertilizers, Steel, cement, electricity. Refinery products (28.04%) have highest weightage and fertilizers (2.63%) have lowest weightage in the Index of Eight Core Industries comprise.

The current weightage, of eight core industries is Petroleum Refinery Products (28.04%) > Electricity (19.85%) > Steel (17.92%) > Coal (10.33%) > Crude Oil (8.98%) > Natural Gas (6.88%) > Cement (5.37%) > Fertiliser (2.63%).

The base year in the current series of ICI is 2011-12.

The Office of the Economic Adviser in the Office of Economic Adviser, Department for Promotion of Industry and Internal Trade, Ministry of Commerce & Industry is responsible for releasing Index of Eight Core Industries.

It is being released with one month time lag (i.e. April Month will be released on 31st May).

➡ **PRODUCER PRICE INDEX (PPI):** It is an index designed to measure the average change in the price of goods and services from the point of view of producer either as they leave the place of production, called Output PPI as they enter the production process, called Input PPI.

PPI is an improvement over WPI, since it is free from the bias of multiple counting inherent in an aggregate commodity basket.

Weight of an item in WPI is based on net traded value defined as value of output adjusted for net imports, whereas in PPI, weights are derived from Supply Use Table (SUT).

A WG on PPI was constituted on 21st August 2014 under the chairmanship of Dr. B N Goldar for PPI.

Now working group of WPI under the chairmanship of Prof. Ramesh Chand, Member, Niti Aayog recommended that the PPI may be released on experimental basis and its weight of 2017-18 may be updated once the MOSPI release the Use Table.

- **Reason:** Strong April-June quarter numbers and expectations that upcoming cuts in Goods and Services Tax (GST) will boost domestic demand.